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United States Patent	12398864
Kind Code	B2
Date of Patent	August 26, 2025
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Holder-and-emitting-member-separable light source module

Abstract

A holder-and-emitting-member-separable light source module, which is a recessed light, a track light or an LED bulb, including a light source holder and a light-emitting member. The light source holder includes a heat dissipation component, a power contact, a circuit contact and a push switch. The light-emitting member includes a plurality of LED chips, a light circuit board, a power management module and a corresponding circuit contact. The light-emitting member is detachably fixed to the light source holder. When the light-emitting member is fixed to the light source holder, the circuit contact is in contact with the corresponding circuit contact. Thus, when an electronic component such as an LED chip or a power module is damaged, only the light-emitting member of the light source module need to be replaced without replacing the light source holder.

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Appl. No.:	18/987461
Filed:	December 19, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20250207764 A1	Jun. 26, 2025

Foreign Application Priority Data

TW	112149797	Dec. 20, 2023
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Publication Classification

Int. Cl.: F21V19/00 (20060101); F21V14/02 (20060101); F21V19/04 (20060101); F21V23/04 (20060101); F21V29/74 (20150101); F21V29/89 (20150101); F21Y115/10 (20160101)

U.S. Cl.:

CPC F21V19/0055 (20130101); F21V14/02 (20130101); F21V19/04 (20130101); F21V23/04 (20130101); F21V29/74 (20150115); F21V29/89 (20150115); F21Y2115/10 (20160801)

Field of Classification Search

CPC: F21V (19/0055); F21V (29/74); F21V (29/89); F21V (14/02); F21V (19/04); F21V (23/04)

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Background/Summary

FIELD OF THE INVENTION

(1) The present invention relates to an LED light source, and more particularly relates to a holder-and-emitting-member-separable light source module.

BACKGROUND OF THE INVENTION

(2) LED light sources have become mainstream in the market due to their low energy consumption and high brightness characteristics. However, LED chips and power modules in the LED light sources are more susceptible to aging and damage than other components of the light source. Due to the non-detachable one-piece design of LED light source products such as LED recessed lights, LED track lights and LED bulbs, users are unable to repair the LED light source by replacing only the LED circuit board when the LED chip or power module is damaged. Therefore, once the LED circuit board is damaged, it can only be replaced together with other components such as heat dissipation elements and circuits, which are extremely durable. This results in significant air pollution caused by the extensive recycling and melting of the LED light source products.

SUMMARY OF THE INVENTION

(3) Accordingly, one objective of the present invention is to provide a holder-and-emitting-member-separable light source module, which allows users to replace only a portion of the light

source rather than the entire light source module when the LED chips, power module, or other electronic components on the light board are damaged.

(4) In order to overcome the technical problems in prior art, the present invention provides a holder-and-emitting-member-separable light source module, comprising: a light source holder, including a heat dissipation component, a power contact, a circuit contact and a push switch, the power contact being provided for connecting to a power source, and an electrical connection between the circuit contact and the power contact being controlled by the push switch; and a light-emitting member, including a plurality of LED chips, a light circuit board, a power management module and a corresponding circuit contact, the plurality of LED chips and the power management module being disposed on the light circuit board and electrically connected to the corresponding circuit contact, the light-emitting member being detachably fixed to the light source holder, wherein when the light-emitting member is fixed to the light source holder, the circuit contact is in contact with the corresponding circuit contact, and the push switch is pressed by the light-emitting member so that the electrical connection is formed between the circuit contact and the power contact, and the plurality of LED chips are electrically connected to the power source through the power management module, the corresponding circuit contact, the circuit contact and the power contact, wherein the holder-and-emitting-member-separable light source module is a recessed light, a track light or an LED bulb.

(5) In one embodiment of the present invention, the holder-and-emitting-member-separable light source module is provided, wherein the heat dissipation component includes heat dissipation fins.

(6) In one embodiment of the present invention, the holder-and-emitting-member-separable light source module is provided, wherein the heat dissipation component is a metal casing.

(7) In one embodiment of the present invention, the holder-and-emitting-member-separable light source module is provided, wherein the light source holder includes a screw portion, the light-emitting member includes a corresponding screw portion, and the corresponding screw portion of the light-emitting member is detachably screwed to the screw portion of the light source holder.

(8) In one embodiment of the present invention, the holder-and-emitting-member-separable light source module is provided, wherein the circuit contact includes two elastic contact elements symmetrical about the screw portion.

(9) In one embodiment of the present invention, the holder-and-emitting-member-separable light source module is provided, wherein the corresponding circuit contact includes inner annular circuit and outer annular circuit concentric around the corresponding screw portion.

(10) In one embodiment of the present invention, the holder-and-emitting-member-separable light source module is provided, wherein the light source holder includes a direction-adjusting bracket.

(11) With the technical means adopted by holder-and-emitting-member-separable light source module of the present invention, the light-emitting member is detachable from the light source holder. Therefore, if any component in the light-emitting member is damaged, the user can replace only the light-emitting member without the necessity of replacing the entire light source, thereby achieving cost savings. Additionally, when the user replaces the light-emitting member, the light source holder can remain in its mounted position, eliminating the inconvenience of demounting and reinstalling the light source holder, making it more suitable for recessed lights which are difficult to demount.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic exploded view illustrating a holder-and-emitting-member-separable light source module according to a first embodiment of the present invention;

(2) FIG. 2 is a schematic exploded side view illustrating the holder-and-emitting-member-separable

light source module according to the first embodiment of the present invention;

(3) FIG. 3 is a schematic side view illustrating the holder-and-emitting-member-separable light source module according to the first embodiment of the present invention;

(4) FIG. 4 is a schematic exploded view illustrating a holder-and-emitting-member-separable light source module according to a second embodiment of the present invention;

(5) FIG. 5 is a schematic exploded side view illustrating a holder-and-emitting-member-separable light source module according to a third embodiment of the present invention; and

(6) FIG. 6 is a schematic exploded side view illustrating a holder-and-emitting-member-separable light source module according to a fourth embodiment of the present invention.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(7) The preferred embodiments of the present invention are described in detail below with reference to FIG. 1 to FIG. 6. The description is used for explaining the embodiments of the present invention only, but not for limiting the scope of the claims.

(8) As shown in FIGS. 1 to 6, holder-and-emitting-member-separable light source modules **100**, **100a**, **100b** according to embodiments of the present invention each comprises: a light source holder **1** and a light-emitting member **2**. The light-emitting member **2** is detachable from the light source holder **1**. Thereby, if any component in the light-emitting member **2** is damaged, the user can replace only the light-emitting member **2** without having to replace the entire light source module, thus achieving cost savings. Additionally, when the user replaces the light-emitting member **2**, the light source holder **1** can remain in its mounted position, eliminating the inconvenience of demounting and reinstalling the light source holder **1**.

(9) The light source holder **1** is detachably mounted at a mounting position of a structure such as a fixed lamp holder, a recessed hole, or a track rail. In the first embodiment as shown in FIGS. 1 to 3, the holder-and-emitting-member-separable light source module **100** is a recessed light, mounted at a ceiling recess hole through a spring clip **16** of the light source holder **1**.

(10) As shown in FIGS. 1 to 3, the light source holder **1** has a heat dissipation component **11**, a power contact **12**, a circuit contact **13** and a push switch **15**. When the user replaces the more easily damaged light-emitting member **2**, the extremely durable light source holder **1** (including the extremely durable heat dissipation component **11**, power contact **12**, circuit contact **13**, and push switch **15**) does not need to be replaced. In this embodiment, the heat dissipation component **11** is a metal casing that efficiently transfers heat generated by the light-emitting member **2** over a wide area by means of good thermal conductivity. The power contact **12** is provided for connecting to a power source for power supply. The circuit contact **13** is electrically connected with the power contact **12** and the circuit contact **13** serves as an electronic contact for contacting the light-emitting member **2**. The electrical connection between the circuit contact **13** and the power contact **12** is controlled by the push switch **15**.

(11) The light-emitting member **2** has a plurality of LED chips **21**, a light circuit board **22**, a power management module **25** and corresponding circuit contacts **23**. The plurality of LED chips **21** and the power management module **25** are disposed on the light circuit board **22** and electrically connected to the corresponding circuit contacts **23**. The power management module **25** is for driving the LED chips **21**. The light-emitting member **2** is detachably fixed to the light source holder **1**. When the light-emitting member **2** is fixed to the light source holder **1**, the circuit contact **13** contacts the corresponding circuit contact **23**, and the push switch **15** is pressed by the light-emitting member **2** so that the electrical connection is formed between the circuit contact **13** and the power contact **12**. At this time, the plurality of LED chips **21** are electrically connected to the power source through the power management module **25**, the corresponding circuit contacts **23**, the circuit contact **13**, and the power contact **12**. When the push switch **15** is not pressed, it causes the electrical connection between the plurality of LED chips **21** and the power source to become an open circuit, thereby preventing the possibility of electric shock.

(12) In this embodiment, the light source holder **1** includes a screw portion **14**, and the light-

emitting member **2** includes a corresponding screw portion **24**. The corresponding screw portion **24** of the light-emitting member **2** is detachably screwed to the screw portion **14** of the light source holder **1**. The screw portion **14** is arranged at the center of a circular plate. The screw portion **14** is a screw tube, and the corresponding screw portion **24** is a screw hole. In other embodiments, the screw portion **14** may be a screw hole, and the corresponding screw portion **24** may be a screw tube. Both configurations allow the light-emitting member **2** to be installed to and demounted from the light source holder **1** in a screwed manner. It goes without saying that the present invention is not limited to this. In other embodiments, the light source holder **1** and light-emitting member **2** may also be fixed in a magnetic, clipping, clamping, or other manner.

(13) In the first embodiment, the circuit contact **13** has two elastic contact elements symmetrical about the screw portion **14**. The elastic contact elements have bent metal pieces, allowing elastic deformation when contacting the corresponding circuit contacts **23**. In other embodiments, the circuit contact **13** may also be at different distances from the screw portion **14** as shown in holder-and-emitting-member-separable light source module **100a** of the second embodiment in FIG. **4**. The corresponding circuit contact **23** comprises two semi-circular circuit paths concentric around the corresponding screw portion **24**.

(14) As shown in FIG. **4**, another difference between the holder-and-emitting-member-separable light source module **100a** according to the second embodiment of the present invention and the holder-and-emitting-member-separable light source module **100** of the first embodiment is that the corresponding circuit contact **23** has an inner annular circuit and an outer annular circuit concentric around the corresponding screw portion **24**.

(15) As shown in FIG. **5**, the holder-and-emitting-member-separable light source module **100b** according to the third embodiment of the present invention is a track light. The light source holder **1** includes a direction-adjusting bracket **17** to adjust the orientation of the LED chips **21**. The heat dissipation component **11** has heat dissipation fins.

(16) As shown in FIG. **6**, the holder-and-emitting-member-separable light source module **100c** according to the fourth embodiment of the present invention is an LED bulb. The heat dissipation component **11** comprises heat dissipation fins.

(17) The above description should be considered as only the discussion of the preferred embodiments of the present invention. However, a person having ordinary skill in the art may make various modifications without deviating from the present invention. Those modifications still fall within the scope of the present invention.

Claims

1. A holder-and-emitting-member-separable light source module, comprising: a light source holder, including a heat dissipation component, a power contact, a circuit contact and a push switch, the power contact being provided for connecting to a power source, and an electrical connection between the circuit contact and the power contact being controlled by the push switch; and a light-emitting member, including a plurality of LED chips, a light circuit board, a power management module and a corresponding circuit contact, the plurality of LED chips and the power management module being disposed on the light circuit board and electrically connected to the corresponding circuit contact, the light-emitting member being detachably fixed to the light source holder, wherein when the light-emitting member is fixed to the light source holder, the circuit contact is in contact with the corresponding circuit contact, and the push switch is pressed by the light-emitting member so that the electrical connection is formed between the circuit contact and the power contact, and the plurality of LED chips are electrically connected to the power source through the power management module, the corresponding circuit contact, the circuit contact and the power contact, wherein the holder-and-emitting-member-separable light source module is a recessed light, a track light or an LED bulb.

2. The holder-and-emitting-member-separable light source module as claimed in claim 1, wherein the heat dissipation component includes heat dissipation fins.
 3. The holder-and-emitting-member-separable light source module as claimed in claim 1, wherein the heat dissipation component is a metal casing.
 4. The holder-and-emitting-member-separable light source module as claimed in claim 1, wherein the light source holder includes a screw portion, the light-emitting member includes a corresponding screw portion, and the corresponding screw portion of the light-emitting member is detachably screwed to the screw portion of the light source holder.
 5. The holder-and-emitting-member-separable light source module as claimed in claim 4, wherein the circuit contact includes two elastic contact elements symmetrical about the screw portion.
 6. The holder-and-emitting-member-separable light source module as claimed in claim 1, wherein the corresponding circuit contact includes inner annular circuit and outer annular circuit concentric around the corresponding screw portion.
 7. The holder-and-emitting-member-separable light source module as claimed in claim 1, wherein the light source holder includes a direction-adjusting bracket.
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